Department of Computer Science & Engineering
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Project Charter
CSE 4316: Senior Design I
Spring 2025

EGG UTA
(Encapsulated Gardening Grounds)

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Revision History

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1 Problem Statement

Many individuals and small-scale farmers struggle to maintain optimal growing conditions for their crops due to factors such as inconsistent watering, improper environmental control, and a lack of real-time data on plant health. These problems are exacerbated in urban settings. Traditional gardening and farming methods often rely on manual observation and intervention, which can be time-consuming and unsustainable for certain individuals. Without an accessible and automated solution, growers may miss out on potential crops and struggle to maintain their land. Addressing this problem with a system that provides real-time monitoring and automation would enhance crop sustainability and make agriculture more accessible to a wider audience, including deployment in urban areas.

2 Methodology

Our team is developing an enclosed gardening system that utilizes sensors, an app, and automated irrigation to assist individuals and farmers in maintaining their crops. The system will continuously monitor key environmental factors such as soil moisture, temperature, humidity, and light levels, providing real-time data through an application. Based on these data, information will be sent to the user so that they can better understand how to care for and maintain the crops contained within. Additionally, this data will drive our automation, such as an automated irrigation system. By integrating smart technology with precision agriculture techniques, our solution will make crop maintenance more accessible, reduce human intervention, and improve overall sustainability in small-scale and indoor farming environments.

3 Value Proposition

Our enclosed gardening system provides significant value to stakeholders by addressing key challenges in sustainable agriculture and smart farming. For industrial sponsors, this project presents an opportunity to explore advances in precision agriculture and automation through the creation of the project. UTA benefits from supporting a project that enables this hands-on, real-world experience. Additionally, our work aligns with broader, traditional goals of sustainability and food security. By investing in this project, stakeholders support the development of accessible, data-driven solutions that can improve crop yields.

4 Development Milestones

- Beginning of Sprint 1 - 7 February 2025
- Beginning of Sprint 2 - 28 February 2025
- Project Charter First Draft - 7 March 2025
- System Requirements Specification (SRS) - 17 March 2025
- Beginning of Sprint 3 - 21 March 2025
- Architectural Design Specification (ADS) - 7 April 2025
- Beginning of Sprint 4 - 11 April 2025
- Begin EGG Implementation - 3 May 2025
- Final Project Demonstration - August 2025

5 Background

For our sponsors, such as UTA, this project represents an investment in innovative agricultural solutions that align with sustainability goals. The team is committed to developing a scalable and accessible system that not only benefits small-scale farmers but also contributes to ongoing research in smart farming techniques. Through this project, we aim to demonstrate the feasibility of data-driven, automated crop maintenance solutions, opening doors for future advancements in sustainable agriculture.

6 Related Work

Innovation and breakthroughs in agricultural processes around the world. The product we aim to create will utilize other technologies in combination to produce benefits in this field. Other technologies similar to this include the following commercially available products.

These products entail some similar concepts to the project we plan to implement. They include self regulating and monitoring for both small-scale personal gardens, and large commercial farms:

- Rain Bird's Drip Irrigation products enable automated irrigation for small-scale farms, allowing for crops to automatically be watered with little human setup or assistance. [1]
- Plantaform's Fogponic System delivers nutrients to plants in a very small garden via a nutrient-rich mist. The product is aimed towards urban customers seeking sustainable gardening solutions. [2]
- Envira's IoT Crop Monitoring Systems offer IoT-based solutions to farmers that enable real-time monitoring of crop health from any device, utilizing data analysis to improve crop efficiency and productivity. [3]
- EOS Data Analytics' Crop Monitoring Software provides a smart agriculture platform that combines satellite data with weather predictions to monitor crop health. [4]

7 System Overview

Our system will consist of an enclosed gardening environment equipped with sensors, automation components, and a user-friendly interface to assist in crop maintenance. The core components will include environmental sensors that monitor key factors such as soil moisture, temperature, humidity, and light levels. This data will be collected and processed in real time, allowing the system to make intelligent decisions about irrigation and environmental adjustments.

Environmental controls will also be implemented, such as automatic irrigation, proper sunlight automation, and ventilation. This will be done with the hardware components needed to achieve this functionality.

Users will interact with the system through a desktop application, which will display real-time data, provide alerts, and allow for manual or automated control of irrigation and environmental settings. The system will track historical trends and optimize future crop growth.

8 Roles & Responsibilities

Product Owner: David Denny
SCRUM Master: Kaitlin Martin
The Team: Khoi Tran, Chanuthi Subasignhage, and Samuel Horn

These roles have been chosen by the group, but may change over time as permitted or needed for the successful development of this project. Currently, these roles are to be permanent. All of these roles have different responsibilities, and are as follows:

The *Product Owner* will be the primary point of communication between sponsors and the team, reaching out when required. The *Product Owner* will also prioritize and adjust features being implemented in the project as needed.
The *SCRUM Master* will be responsible for ensuring SCRUM deadlines are met on time, ensuring the team is functioning properly and efficiently. This includes communicating with all team members to make sure they are successfully filling all of their required responsibilities. This includes communication through scheduled meetings and Discord. Additionally, our team will be utilizing a Notion workspace, and it will be overseen by the group and *SCRUM Master*.
Everyone on the *the team*, including the *Product Owner* and *SCRUM Master* are responsible for the development of the project. *The team* is self-organizing, and will meet and work on the project as needed to achieve sprint requirements.

9 Cost Proposal

The primary source of funding for this project is obtained from the University. The cost of our project must take into consideration all of the hardware and software needed to see the project come to fruition. On the hardware level, multiple sensors, a pump, and a microcontroller to drive these peripherals. On the software level, the costs of hosting a web application must be considered. We are starting with an initial budget of $800.00 from UTA.

9.1 Preliminary Budget

Below is a high level look at the estimated costs of equipment and software needed for this project:

Equipment / Software Needed	**Estimated Price Range**
Web Hosting Services (AWS)	$10.00 - $20.00
Domain Name	$10.00 - $20.00
Database (AWS)	$15.00 - $50.00
Microcontroller (Arduino/Raspberry Pi)	$20.00 - $100.00
Sensors (Humidity/Temperature/Light Level/etc.)	$50.00 - $100.00
Pump	$10.00 - $20.00
Enclosure Materials (Base/Glass/Hardware and Sensor Mounting/etc.)	$100.00 - $200.00
Soil and Plants	$25.00 - $50.00

Table 2: Estimated Project Expenditures

9.2 Current & Pending Support

Currently, the total budget we have for this project is limited to the intial $800.00 that the CSE department at UTA provides. We are open to the idea of finding outside sponsorships and scholarships to find funding that could enhance our project, or as needed.

10 Facilities & Equipment

As needed, the various labs within the ERB will be utilized to create this project. Meetings will similarly be held there. For the assembly of the hardware used in this project, we will use various tools already present within the lab. Depending on the enclosure we use, makerspaces may be required, in addition to any 3D-printed parts we may implement. Other tools to implement the project (both the Software and Hardware) will be acquired as needed.

11 Assumptions

Below is a list of critical assumptions held by our team. These include the items and processes we assume to be true in regards to our projects creation, including provided equipment:

- The project will involve the development of a controlled, enclosed environment optimized for plant growth.
- Implementation of a mobile application to monitor and display data collected from the greenhouse.
- There will be an automated irrigation system.
- Parts and sensors will be available within certain deadlines.
- Weather for testing our agricultural project will be ideal.
- Workspace allowing for the use of soil and water will be available.
- Hardware will be available and operational soon enough to allow for time to implement data analysis.

12 Constraints

The following list contains key constraints related to the implementation and testing of this project:

- Final prototype demonstration must be completed by the end of the Summer Semester in August of 2025.
- The implementation of this project must satisfy all ABET requirements for this course.
- Total development costs must not exceed $800.00. All purchased hardware and software must adhere to this budget.
- All implemented devices and environments must adhere to agreed upon standards.

13 Risks

Risk description	Probability	Loss (days)	Exposure (days)
Availability of X sensor module due to contractor delay	0.50	20	10
Outdoor testing grounds are not available	0.50	14	7.0
Availability of a space to work with water and soil	0.50	10	5.0
Unexpected budget constraints	0.10	20	2.0
Non-ideal environmental factors for growing	0.10	20	2.0
Data collection does not begin as soon as expected	0.10	20	2.0

Table 3: Overview of highest exposure project risks

14 Documentation & Reporting

14.1 Major Documentation Deliverables

14.1.1 Project Charter

The Project Charter is maintained using Overleaf so that the team can collaborate on the document. The document may be updated at the discretion of any individual, and changes can be discussed and agreed upon in team meetings. The initial version will be delivered on March 7th, 2025. The final version of this document will be finalized and delivered by the specified demonstration day in August of 2025.

14.1.2 System Requirements Specification

Similarly to the Project Charter, the System Requirements Specification (SRS) will be created in Overleaf and maintained as needed for the goals of the project. This document will contain the list of requirements we will work towards during each sprint. During the course of the project's creation, the scope of the project may increase or decrease. The SRS will change and be updated to reflect those changes in scope as needed. The initial version will be delivered on March 17th, 2025. The final version of this document will be delivered by the specified demonstration day in August of 2025.

14.1.3 Architectural Design Specification

The Architectural Design Specification (ADS) will be created in Overleaf and maintained as needed for the goals of the project. During the course of the project's creation, the scope of the project may increase or decrease. The ADS will change and be update to reflect those changes in scope as needed. The initial version will be delivered on April 7th, 2025. The final version of this document will be finalized and delivered by the specified demonstration day in August of 2025.

14.1.4 Detailed Design Specification

The Detailed Design Specification (DDS) will be created in Overleaf and maintained as needed for the goals of the project. During the course of the project's creation, the scope of the project may increase or decrease. The DDS will change and be updated to reflect those changes in scope as needed. It is unknown when the initial version will be delivered, but we estimate late April/early May. The final version of this document will be finalized and delivered by the specified demonstration day in August of 2025.

14.2 Recurring Sprint Items

14.2.1 Product Backlog

The product backlog reflects the requirements contained in the SRS. Each backlog for each sprint will assign requirements from the backlog to work on for the upcoming sprint. Through group assignments and team leadership decisions, requirements and tasks from the backlog will be spread out among team members to work on for the upcoming sprint. This selection will be finalized during the group meeting held just before the start of each sprint. Items completed during a sprint will be removed from the backlog, and the product backlog will reflect this.

14.2.2 Sprint Planning

As mentioned, meetings will be held regularly each week (most often, these meetings will be held Tuesdays and Fridays). Both the team and leadership will decide on the items in the Product Backlog and SRS to work on for the upcoming sprint. Additionally, these tasks will be assigned to various team members at the team's discretion.

14.2.3 Sprint Goal

Both the team and leadership will decide on the items in the Product Backlog and SRS to work on for the upcoming sprint. Additionally, these tasks will be assigned to various team members at the team's discretion. In this way, the items from the backlog and SRS represent the goals for the upcoming sprint.

14.2.4 Sprint Backlog

The sprint backlog will be a list of the items and tasks from the SRS and Product Backlog that will be worked on during the current or upcoming sprint. As mentioned before, both the team and leadership will decide on the items to go into the Sprint Backlog to work on for the upcoming sprint.

14.2.5 Task Breakdown

As mentioned before, both the team and leadership will decide on the items in the Product Backlog and SRS to work on for the upcoming sprint. Additionally, these tasks will be assigned to various team members at the team's discretion.

14.2.6 Sprint Retrospective

Our team uses notion to keep track of our sprint backlogs, which we can use to prioritize and categorize tasks by difficulty. This discussion generally occurs after class on Tuesdays and Thursdays as a check-in after sprints.

14.2.7 Individual Status Reports

Individual Status Reports will be created and submitted by each member of the team following the end of a sprint. These reports will follow the given template, and ensure each member is held responsible for the work they do during each sprint. If any problems should arise in the creation of this project, these reports may be referenced understand and fix them.

14.2.8 Engineering Notebooks

This team does not utilize engineering notebooks for the project.

14.3 Closeout Materials

14.3.1 System Prototype

The final system project will be the culmination of the entire project, including everything created with hardware and software. The web deliverables and physical garden will be demonstrated in tandem with

each other as needed on demonstration day in August of 2025. This demonstration will be performed in one of the requisite labs.

14.3.2 Project Poster

The project poster will be a larger poster summarizing the function and operation of our project. It will include details on how a user may interact and use the gardening grounds. It will also include the specifications of the software and hardware deliverables used in the project's creation. These qualities will display the concepts relating to our degrees and requirements that are satisfied in it's implementation.

14.3.3 Demo Video

Once finished, the EGG will be shown in a demo video that contains both the encapsulated system and user interaction through the app or webpage. This video will show how the garden reacts to user commands and how they can access real-time data on the plants in the garden. A time-lapse could be utilized to show how plants thrive in the EGG.

14.3.4 Source Code

All created source code will be provided to potential users. These source files will also be submitted as final deliverables upon the project's completion. Licensing terms will be decided and agreed to by the team by end of the projects implementation, which will be included where it is deemed necessary.

14.3.5 Source Code Documentation

All created source code will be documented and submitted as a final deliverable upon the project's completion. This documentation will follow the format that best works with the languages we will use.

14.3.6 Hardware Schematics

As used in the project's designs, simple wiring diagrams showing hardware device connections will be included and submitted as a final deliverable. Programs such as LTSpice will be used to accomplish this.

14.3.7 CAD files

In the event that CAD modeling is used (likely for the enclosure's creation), the files created for the project's implementation will be maintained and submitted as a final deliverable. CAD software likely to be used in this project would include programs such as Solidworks or Blender.

14.3.8 User Manual

In addition to the project poster giving a brief overview of the device's operation, a simple User Manual will be created, which will give a more detailed explanation of the systems used in the projects design, and how an uninformed user may learn to operate the device.

14.3.9 Installation Scripts

Explanations for the installation of required software will be discussed in the user manual mentioned previously.

References

[1] R. B. Corporation, "Drip irrigation and low-volume irrigation systems," 2025.

[2] N. Y. Times, "How AI and Nasa can help gardeners lacking a green thumb," 2025.

[3] Envira, "IoT sensors to monitor crop health," 2025.

[4] E. D. Analytics, "Crop Monitoring Platform For Remote Farm Risk Management," 2025.